

CO3-AT3-Assignment-2

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QUESTIONS

Q1
Food Ordering System ♦ Multiple Threads
25 marks

Q2
Multithreaded Number Processing
25 marks

Q3
Producer-Consumer ♦ Inter-Thread Communication
25 marks

Q4
Online Examination ♦ Thread Synchronization
25 marks

3/4 answered

Q1 Food Ordering System ♦ Multiple Threads

25 marks

Problem Statement:

Develop a multithreaded Java program for a food ordering system. Separate threads should process order placement food preparation and delivery.

Requirements:

Create three separate threads.

Each thread should perform its respective task.

Display the execution status of each task.

Demonstrate concurrent execution.

Your Code

```
1 public class Main {
2     static class SharedBuffer {
3         private final int[] buffer;
4         private int count = 0, front = 0, rear = 0;
5         SharedBuffer(int size) {
6             buffer = new int[size];
7         }
8         public synchronized void put(int item) {
9             while (count == buffer.length) {
10                 try { wait(); } catch (InterruptedException e) {}
11             }
12             buffer[rear] = item;
13             rear = (rear + 1) % buffer.length;
14             count++;
15             System.out.println("Produced: " + item + " (count=" + count);
16             notify();
17         }
18     }
19 }
```

Input

stdin input

Output

```
Producer Consumer started
Produced: 1 (count=1)
Consumed: 1 (count=0)
Produced: 2 (count=1)
Consumed: 2 (count=0)
Produced: 3 (count=1)
```

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Q4

Online Examination ♦ Thread
Synchronization
25 marks

4/4 answered

Q2 Multithreaded Number Processing

25 marks

Problem Statement:

Develop a Java program that creates multiple threads to perform different operations on a set of numbers.

Requirements:

Create separate threads to calculate the sum average and maximum value.

Use the Thread class.

Execute the threads concurrently.

Display the result produced by each thread

Your Code

```
1 public class Main {  
2     static class SumThread extends Thread {  
3         private int[] numbers;  
4         private int sum;  
5         SumThread(int[] numbers) {  
6             this.numbers = numbers;  
7         }  
8         public int getSum() {  
9             return sum;  
10        }  
11        public void run() {  
12            sum = 0;  
13            for (int num : numbers) {  
14                sum += num;  
15            }  
16            System.out.println("Sum calculated: " + sum);  
17        }  
18    }
```

Input

stdin input

Output

```
Starting Multithreaded Number  
Processing  
Max calculated: 40  
Sum calculated: 95  
Average calculated: 19.0
```

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Thread Synchronization
25 marks

3/4 answered

Q3 Producer-Consumer Inter-Thread Communication

25 marks

Problem Statement:

Develop a Java program to implement a Producer-Consumer scenario using inter-thread communication.

Requirements:

Create a producer thread and a consumer thread.

Use a shared buffer.

Use wait() and notify() for communication.

Ensure that the consumer waits when the buffer is empty.

Ensure that the producer waits when the buffer is full.

Your Code

```
1 public class Main {  
2     static class SharedBuffer {  
3         private final int[] buffer;  
4         private int count = 0;  
5         private int front = 0;  
6         private int rear = 0;  
7         public SharedBuffer(int size) {  
8             buffer = new int[size];  
9         }  
10        public synchronized void put(int item) {  
11            while (count == buffer.length) {  
12                try {  
13                    wait();  
14                } catch (InterruptedException e) {  
15                    System.out.println("Producer interrupted.");  
16                }  
17            }  
18        }  
19    }  
20 }
```

Input

stdin input

Output

```
Producer Consumer started  
Producer produced: 1 | Buffer  
count = 1  
Consumer consumed: 1 | Buffer  
count = 0  
Producer produced: 2 | Buffer
```

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Online Examination ♦ Thread
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25 marks

3/4 answered

Q4 Online Examination ♦ Thread Synchronization

25 marks

Problem Statement:

Develop a Java program to simulate an online examination system where multiple students submit answers simultaneously.

Requirements:

Create multiple student threads.

Maintain a shared submission counter.

Use synchronization while updating the counter.

Display each student's submission status.

Display the final number of submissions.

Your Code

```
1 public class Main {  
2     private int submissionCounter = 0;  
3     public synchronized void submitAnswer(String studentName) {  
4         System.out.println(studentName + " is submitting answer");  
5         try {  
6             Thread.sleep(100);  
7         } catch (InterruptedException e) {  
8             }  
9         submissionCounter++;  
10        System.out.println(studentName + " submitted successfully. Total submissions = " + submissionCounter);  
11    }  
12    public int getFinalCount() {  
13        return submissionCounter;  
14    }  
15    static class StudentThread extends Thread {  
16        private final Main system;  
17        private final String studentName;
```

Input

stdin input

Output

```
Student_A is submitting answer  
Student_A submitted successfully.  
Total submissions = 1  
Student_E is submitting answer  
Student_E submitted successfully.  
Total submissions = 2
```